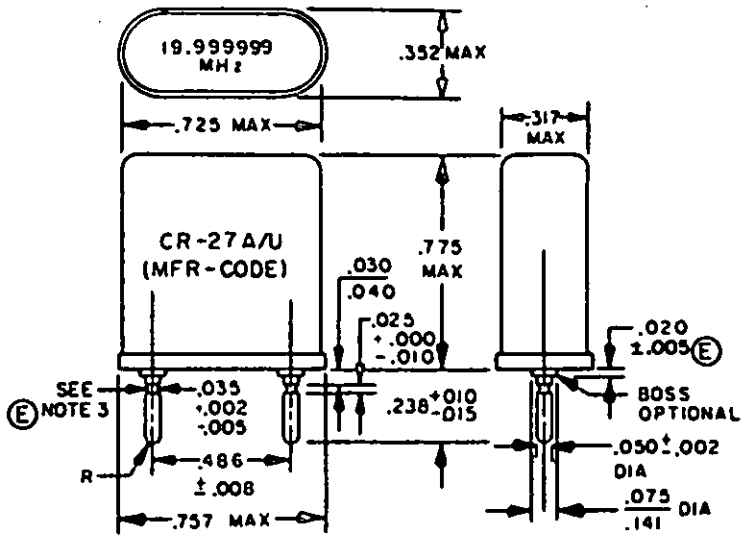
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-27A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
E 3. The pin undercut may be omitted.

E denotes changes.

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	0.8 to 20 MHz, inclusive
Capacitance, shunt	7 pF maximum
Frequency tolerance, operating temperature range	±20 PPM
Frequency stability	± 5 PPM
Equivalent resistance	See table II
Antiresonance, load capacitance	32.0 ±0.5 pF
Mode of oscillation	Fundamental
Reference temperature	75° ±1°C
Temperature ranges:	
Operating	70° to 80°C, inclusive
Operable	-55° to +70°C and 80° to 90°C, inclusive
Test set, calibration values, and rated drive level	See table III
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

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TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	620
0.85 + to 0.9	600
0.9+ to 1	570
1+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.5	410
1.5+ to 1.62	370
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2	290
2+ to 2.12	270
2.12+ to 2.25	240
2.25+ to 2.6	190
2.6+ to 3	150
3+ to 3.4	110
3.4+ to 3.75	90
3.75+ to 4	75
4+ to 5	60
5+ to 7	35
7+ to 10	24
10+ to 15	22
15+ to 20	20

TABLE III. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	ohms	mA	volt	mW
TS-330/TSM	0.8 to 1.5	0	1/	---	5.0 ±1.0 2/
	1.5+ to 2.25	50	10	---	5.0 ±1.0
	2.25+ to 3.4	50	10	---	5.0 ±1.0
	3.4+ to 5.1	22	15	---	5.0 ±1.0
	5.1+ to 7.5	13	20	---	5.0 ±1.0
	7.5+ to 10	13	20	---	5.0 ±1.0
	10+ to 15	11	15	---	2.5 ±0.5
TS-683/TSM	15+ to 20	10	---	0.16	2.5 ±0.5

1/ Set crystal current control at *extreme counterclockwise (minimum)* position.
2/ The violation of $P = I^2R$ is intentional; at the higher resistance of the crystal unit, the empirical power dissipation will be as rated.

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/4 (Sockets, Crystal, Top and Bottom Mounting, 2 Contacts). For Navy use, other sockets listed in MIL-STD-242 may be used.

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Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, OS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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